

Simulated Touch: A Meta-Analysis of the Effects of Virtual Social Touch on Interpersonal Communication Outcomes

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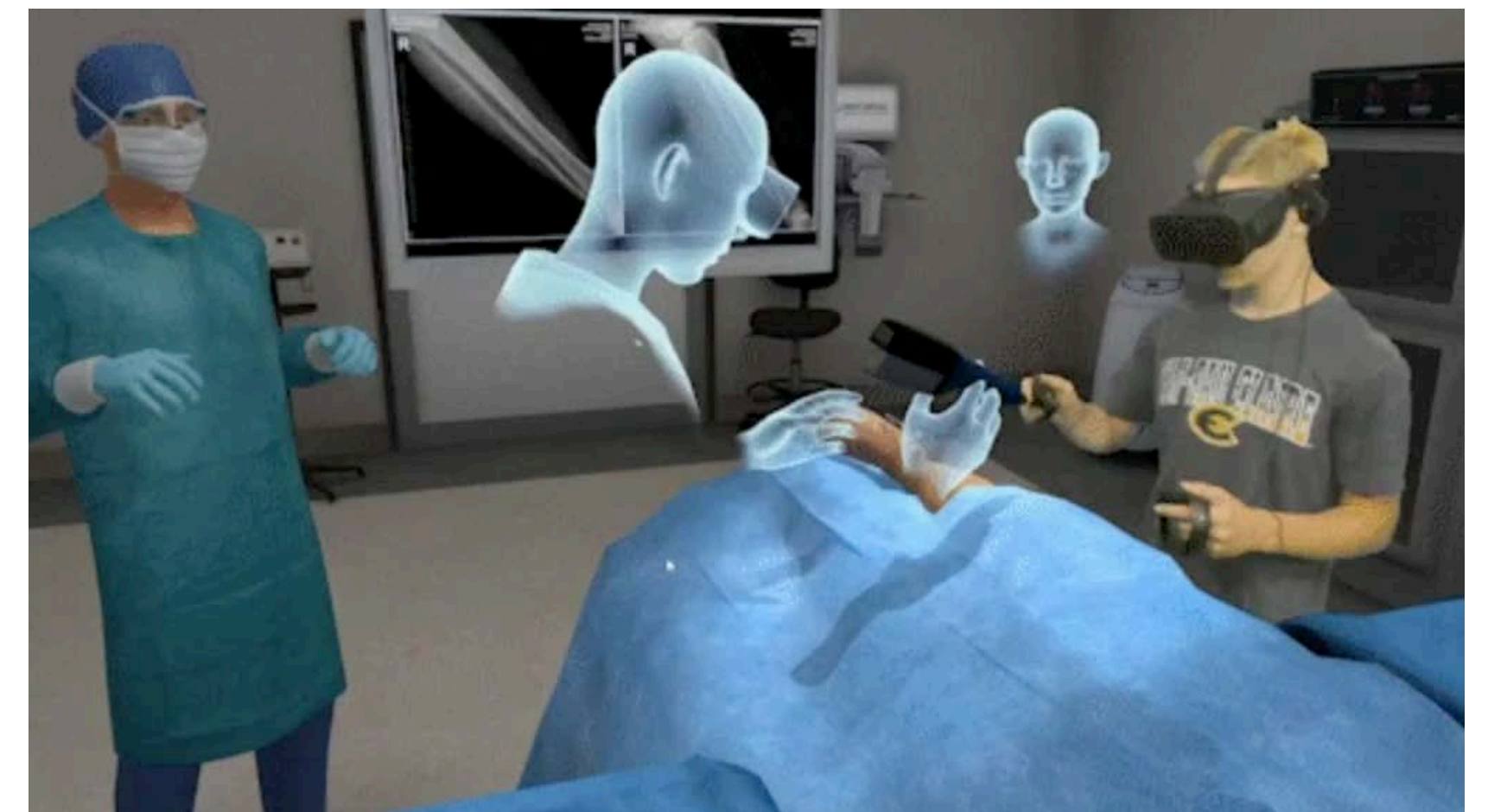
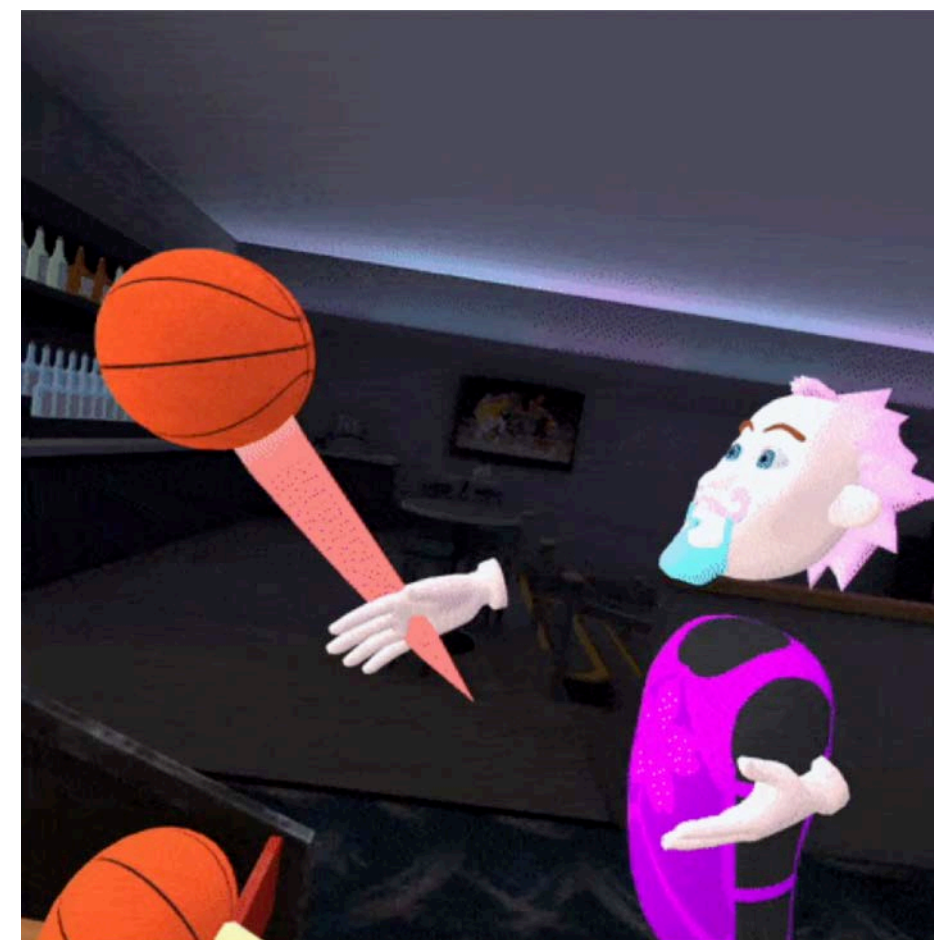


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Why Should We Care About Social Touch in VEs?

- One interesting topic in the domain of virtual environments (VEs) pertains to the investigation of social interactions and broader interactive abilities. Current VEs are increasingly replicating the real world in sophisticated ways, leading to a notable **focus on artificially reproducing human senses** in social interactions.
- In light of the advancement of haptic technologies, it is expected to surmount the technical challenges that currently impede the realistic simulation of human touch. Consequently, there has been a significant amount of scholarly interest and individual studies focused on social touch during interpersonal communication in VEs.



Why Should We Care About Social Touch in VEs?

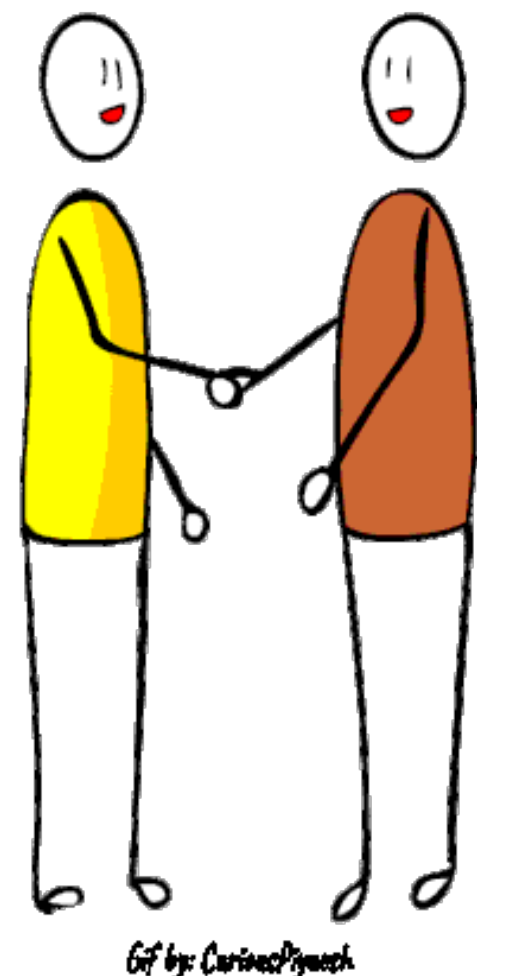
- However, despite this attention, scholarly inquiry in this domain remains fragmented. For example, effects of social touch (e.g., “Midas effect”) widely accepted in the physical world yield similar outcomes in VEs, yet they may also lack empirical support or even produce contradictory results.
- Thus, understanding **how virtual social touch (VST) reproduced in VEs differs from real social touch** and **how these differences impact human interactions** remains a crucial research inquiry. The purpose of the current research is to systematically synthesize studies on the effects of VST within a virtual communication context through meta-analysis.

Research Questions are formulated as follows:

- Research Question1: Is there an effect of virtual social touch on interpersonal communication outcome?
- Research Question2: What are the factors that moderate the effect of virtual social touch?

Outcomes of Interpersonal Social Touch

- Social touch is defined from two perspectives (Gliga et al., 2019; Saarinen et al., 2021):
 - (a) in terms of sensory qualities of touch (i.e., "CT-optimal touch")
 - (b) in terms of interpersonal and intentional aspects of touch (i.e., affective signals conveyed by touch)
- Most discussions around social touch (e.g., Field, 2019; Hertenstein et al., 2006; Huisman, 2017) have primarily focused on its contributions to physical and emotional wellbeing, attachment and bonding, changes in attitudes and behaviors, and the communication of affect.

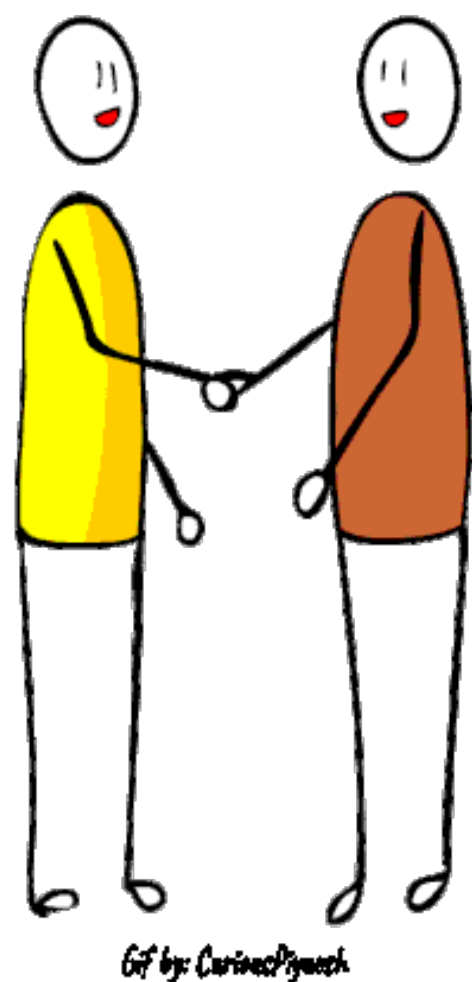


Outcomes of Interpersonal Social Touch

- To assess the magnitude of the effects of VST on various outcomes, outcomes measures for current study was categorized and analyzed based on perceptual, affective, behavioral, and physiological aspects.

Category	Outcome measures included in current study
Perceptual	• Evaluations of interaction (e.g., Overall experience, Emotional contagion) • Evaluations of partner (e.g., Liking, Trust, Positive evaluation, Impression, Perceived closeness)
Affective	• Subjective feelings (e.g., Pleasantness, Excitement, Amusement)
Behavioral	• Amount of interaction (e.g., Duration (Time), Volume of exchanged messages) • Compliance/Acceptance with request to aggress
Physiological	• Arousal responses (e.g., Heart Rate, Skin Conductance Response (SCR))

Note. This classification is based on the categorization by Van Erp & Toet (2015) and Sarrinen et al. (2021).



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Factors Influencing Virtual Social Touch

- **Demographic factors**, such as gender, age, and cultural background, are among the primary influences on social touch (Hertenstein et al., 2006).
- Also, Saarinen et al. (2021) identified two primary contextual factors that can modulate an individual's immediate responses to social touch through a systematic review of empirical studies on the social touch.
 - **(a) Toucher's characteristics:** Prejudices and group membership of toucher (e.g., ethnicity, political, or sexual orientation), Quality of relationship with target person (familiar, stranger, partner etc.), Dynamic facial expressions, Physical attractiveness
 - **(b) Situational factors:** Attentiveness to such exposure, Multimodal environment (e.g., orders, voices), Target person's exposure to sensory pain or psychological distress

Factors Influencing Virtual Social Touch

- We coded variables that could moderate the VST effect, including demographic, contextual, and experimental factors.

Category	Moderators included in current study
Demographic factor	• Gender (coded as gender distribution of female among samples) • Age (coded as mean age of samples; continuous) • Cultural background (coded as geographic location, i.e., continents)
Contextual factor	• Type of interaction context (coded as (a) private or (b) neutral) • Type of toucher (coded as (a) human-controlled agent or (b) computer-controlled agent)
Experimental factor	• Type of touch (coded as (a) vibrotactile, (b) thermal, (c) force, and (d) contactless)
Other factor	• Year of publication (coded as year the study was published; continuous) • Type of sample (coded as (a) student or (b) adults) • Publication status (coded as (a) published (i.e., peer-reviewed journal articles or conference paper) or (b) unpublished (i.e., dissertations or unpublished data)).

Note. Studies in which gender distribution was not reported were excluded from the moderator analysis. Studies in which geographic location was not specified were considered to have been conducted at the authors' location of operation. The factors of sample type and publication status were excluded from the final moderator analysis, as the majority of studies were published and utilized student samples (90% each).

Literature Search

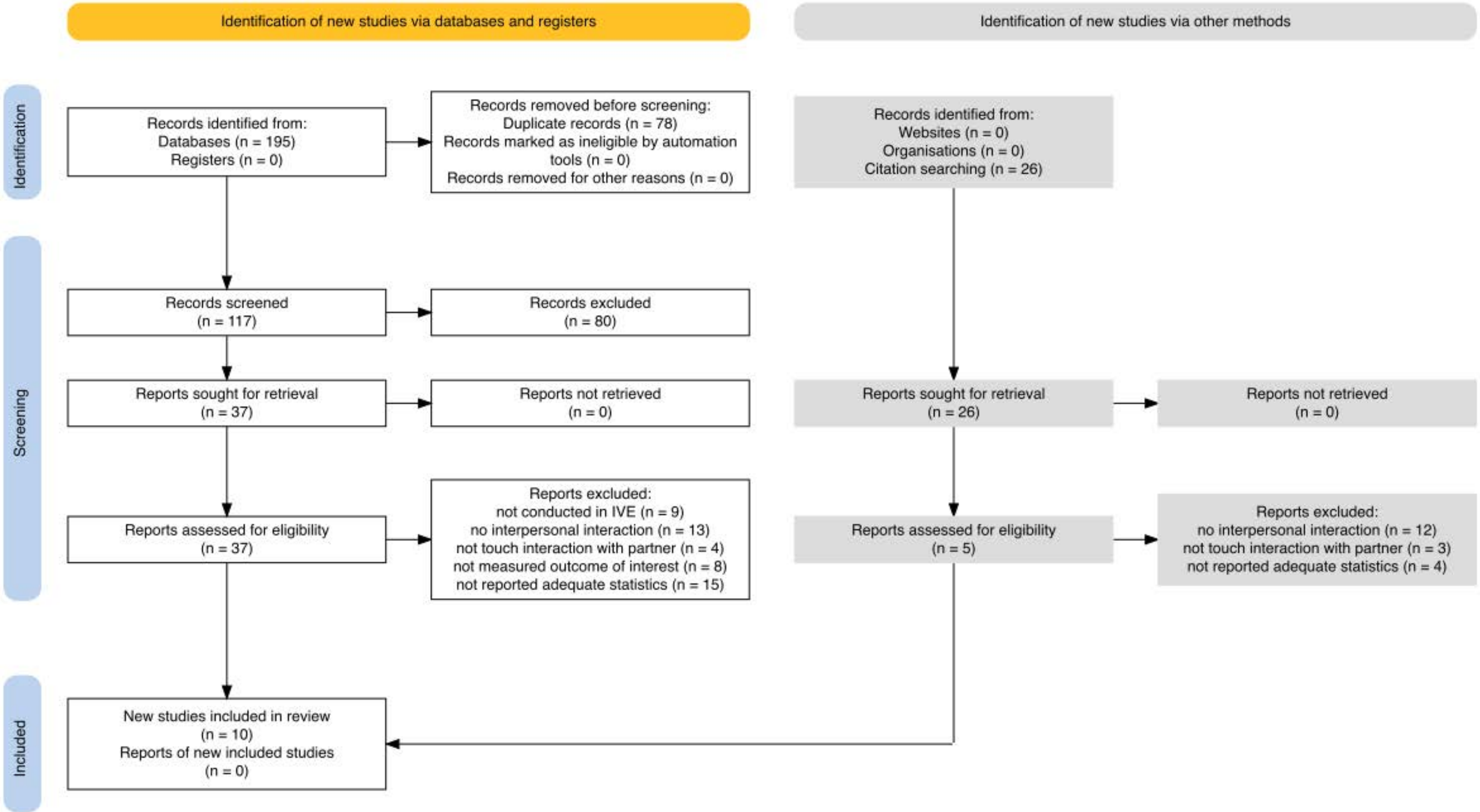
- The current meta-analysis adheres rigorously to the guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA) statement (Page et al., 2021).
- We analyzed studies published in peer-reviewed journals, and then included a conference paper that satisfies the inclusion criterion because there were not many studies suitable for meta-analysis in the process of collecting literature.
- In April 2024, a systematic search was undertaken in Pubmed, Scopus, Web of Science, and ProQuest.
- We included studies that contained words related to virtual settings and interpersonal touch in the title or abstract.
- The general search across all databases used the following logic: ("virtual reality" OR "VR" OR "augmented reality" OR "AR" OR "mixed reality" OR "MR" OR "virtual environment") AND ("virtual touch" OR "social touch" OR "interpersonal touch" OR "mediated touch" OR "affective touch" OR "simulated touch" OR "haptic communication").

Inclusion Criteria

We identified studies satisfying the following criteria:

- First, the studies must have been conducted within interaction contexts utilizing immersive VE technology.
 - Second, the studies must involve interpersonal interaction contexts with a partner.
 - Third, the studies must involve touch with interaction agents.
 - Fourth, the studies must measure interaction outcomes resulting from exposure to VST.
 - Fifth, the studies must report adequate statistical data for calculating an effect size (e.g., Ms and SDs/SEs, *t* statistics).
 - Sixth, the studies must be written in English.
-
- Based on the above inclusion criteria, we included 10 studies with 12 samples, 48 effect sizes, and 481 participants.

Inclusion Criteria



Assessment of Study Quality

- Following “APA Publications and Communications Board Working Group on Journal Article Reporting Standards”¹⁾, we assessed the overall quality of our included studies across eight themes outlined in the *Reporting Standards for Studies With an Experimental Manipulation or Intervention*: Experimental manipulations or interventions, Units of delivery and analysis, Participant flow, Treatment fidelity, Baseline data, Statistics and data analysis, Adverse events and side effects, and Discussion.
- Each study was rated as (a) Poor (if any specific descriptions corresponding to the standards were not reported), (b) Fair (if some details were reported), or (c) Good (if most details were reported).
- Each individual study was considered to meet a certain level of quality as none received poor ratings for two or more of the eight standard items. Also, 60% of the included studies reported obtaining Institutional Review Board approval.

¹⁾ DOI:10.1037/0003-066X.63.9.839

Meta-Analytic Strategy

Calculation of Effect Sizes and Its Variances

- Eligible articles were coded, and an effect size was calculated for each outcome measure.
- Effect sizes were computed as $(M_{\text{treat}} - M_{\text{control}}) / SD_{\text{pooled}}$ for studies reporting group comparisons ($n=7$), following the formula in Lipsey and Wilson (2001).
- For studies measuring a binary dependent variable (e.g., Acceptance (%), Yes or No questions), effect sizes were calculated by converting $\log(\text{OR})$ to *Cohen's d* ($n=2$).
- For studies lacking sufficient data to directly calculate effect sizes from means and standard deviations or proportions (e.g., when standard deviations were not reported, or only test statistics were provided), effect sizes were derived from p -values from Wilcoxon signed rank test ($n=1$).

Meta-Analytic Strategy

Inclusion Bias Analyses

- In line with contemporary best practices for reducing inclusion bias, current study employed a **RoBMA-PSMA (Robust Bayesian Meta-Analysis Publication Selection Model-Averaging)**¹⁾ approach (Bartoš et al., 2022; Bartoš et al., 2023).
- Frequentist methods have general limitations, including (a) ignoring model uncertainty, (b) inability to quantify evidence for the absence of an effect, and (c) dependence on an unknown sampling plan (Bartoš et al., 2023; Hinde et al., 2020; Maier et al., 2023). Additionally, the meta-analytic mean effect size is often overestimated or cannot be estimated with small sample sizes (i.e., “small-study effects”).
- Therefore, given the limited size of interpersonal communication studies conducted in IVEs, the current study adopts this method to enable flexible adaptation to scenarios with high heterogeneity and small sample sizes.

¹⁾ Recent simulation studies comparing the performance of 13 publication bias correction methods (e.g., PET-PEESE, selection models, p-curve, WAAP-WLS) in terms of various metrics (e.g., FPR, FNR, OF, RMSE) have demonstrated that RoBMA-PSMA generally outperforms other methods.

Meta-Analytic Strategy

Inclusion Bias Analyses

Limitations of PET-PEESE ¹⁾ and Selection models ²⁾	RoBMA's benefits
• Frequentist Neyman-Pearson point-null hypothesis significance tests (NHSTs) are based on binary accept/reject decisions.	• BF allows researchers to quantify relative evidence for the null hypothesis and thus distinguish between absence of evidence and evidence of absence. • The prior distributions allow the selection models to be estimated even in cases with few p-values in some of the p-value intervals
• Accumulation bias is inevitable when using a frequentist method.	• BFs allow for sequential updating, meaning that the analysis can be updated without having to worry about accumulation bias.
• PET-PEESE and selection models will sometimes arrive at different results.	• The model averaging obviates the need to select a single model in an all-or-none fashion.

Note. The author has summarized the methodological strengths of RoBMA as presented in Bartoš et al. (2022).

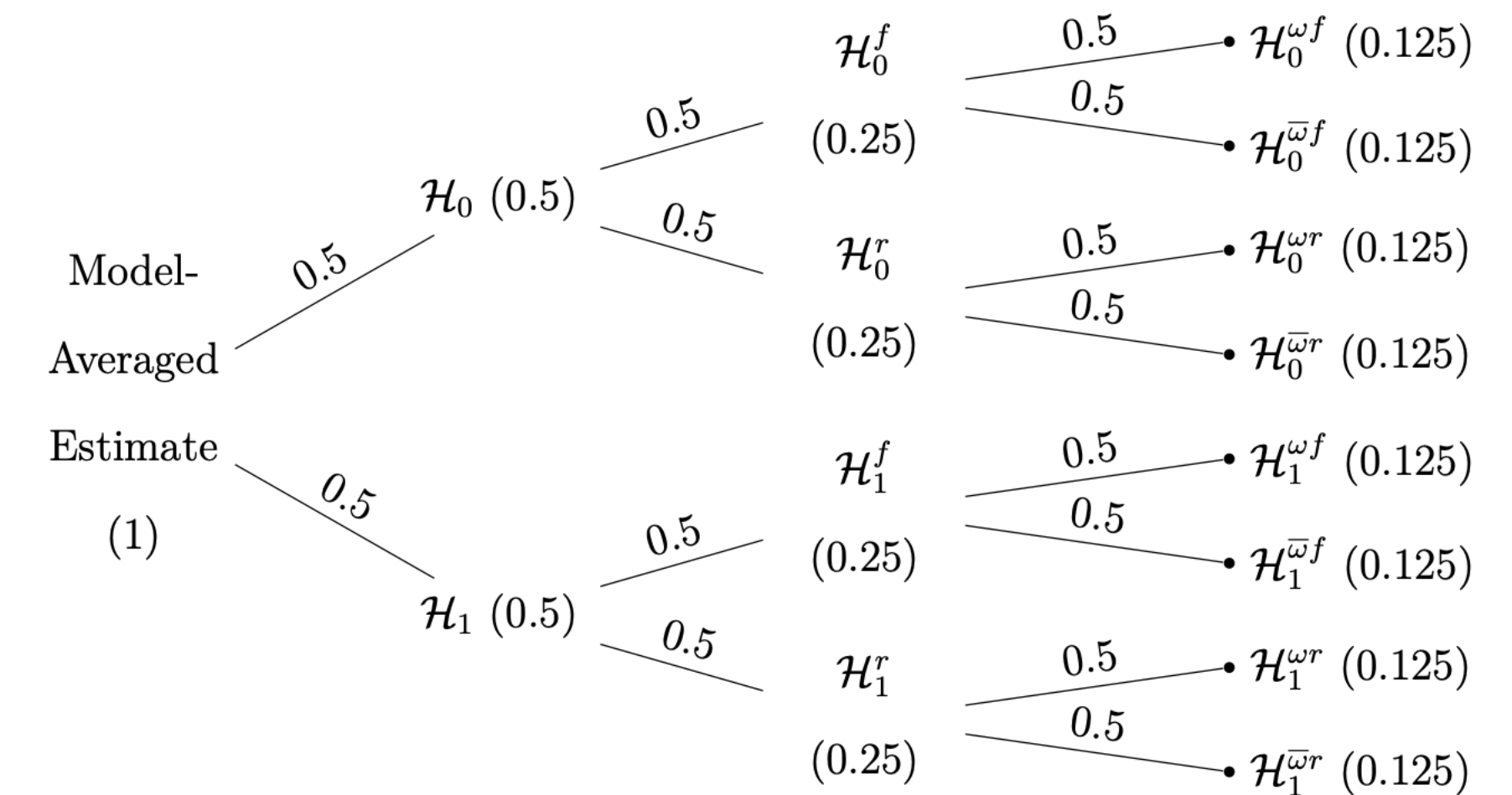
¹⁾ PET-PEESE: publication-bias-adjustment method that corrects for the correlation between effect size and standard errors or effect sizes and standard errors squared

²⁾ Selection model: publication-bias-adjustment methods that use weighted likelihood to account for studies that are missing because of publication bias

Meta-Analytic Strategy

Inclusion Bias Analyses: specification of prior distributions

- Effect size (μ) : *Student* – $t_{[0,\infty]}(0.35,0.102,3)$, i.e., "Oosterwijk prior", an informed prior distribution for small- to-medium-sized effects
- Heterogeneity (τ) : *InvGamma*(1,0.15), empirical prior distribution based on heterogeneity estimates recorded from meta-analyses
- Publication bias regression coefficients (β) : *Cauchy* $_{[0,\infty]}(0,1)$ for PET and *Cauchy* $_{[0,\infty]}(0,5)$ for PEESE
- Publication bias weights (ω) : Simulation-Based Prior Distribution
 - $\omega_{Two-sided(.05)} \sim CumDirichlet(2.49,0.83)$
 - $\omega_{Two-sided(.1,.05)} \sim CumDirichlet(2.88,0.98,0.99)$
 - $\omega_{One-sided(.05)} \sim CumDirichlet(2.61,0.89)$
 - $\omega_{One-sided(.05,.025)} \sim CumDirichlet(2.92,0.95,0.75)$
 - $\omega_{One-sided(.5,.05)} \sim CumDirichlet(3.17,0.80,0.83)$
 - $\omega_{One-sided(.5,.05,.025)} \sim CumDirichlet(3.24,1.02,0.68,0.66)$



* adopted from Maier et al. (2023)

Meta-Analytic Strategy

Moderator analyses

- The existing meta-regression methods have been criticized for their reliance on frequentist testing and p -values, making interpretation challenging for researchers (Wagenmakers, 2007), as well as for not considering the uncertainty of publication bias, which can lead to overestimation (Bartoš et al., 2023; Maier et al., 2023).
- Thus, to assess the effect of moderators of interest in current study, RoBMA-regression analysis was conducted.¹⁾
- The influence of each moderator was assessed using visualization plots of the marginal model-averaged prior and posterior distributions of the RoBMA-regression models, respectively.

¹⁾ According to recommendations by Bartoš et al. (2023), $\mu \sim \text{Normal}(0,0.25)$ prior distribution was specified for categorical moderators, while a $\beta \sim \text{Normal}(0,0.25)$ prior distribution was specified for continuous moderators, corresponding to an expectation of mostly small to medium moderation effects.

Study Characteristics

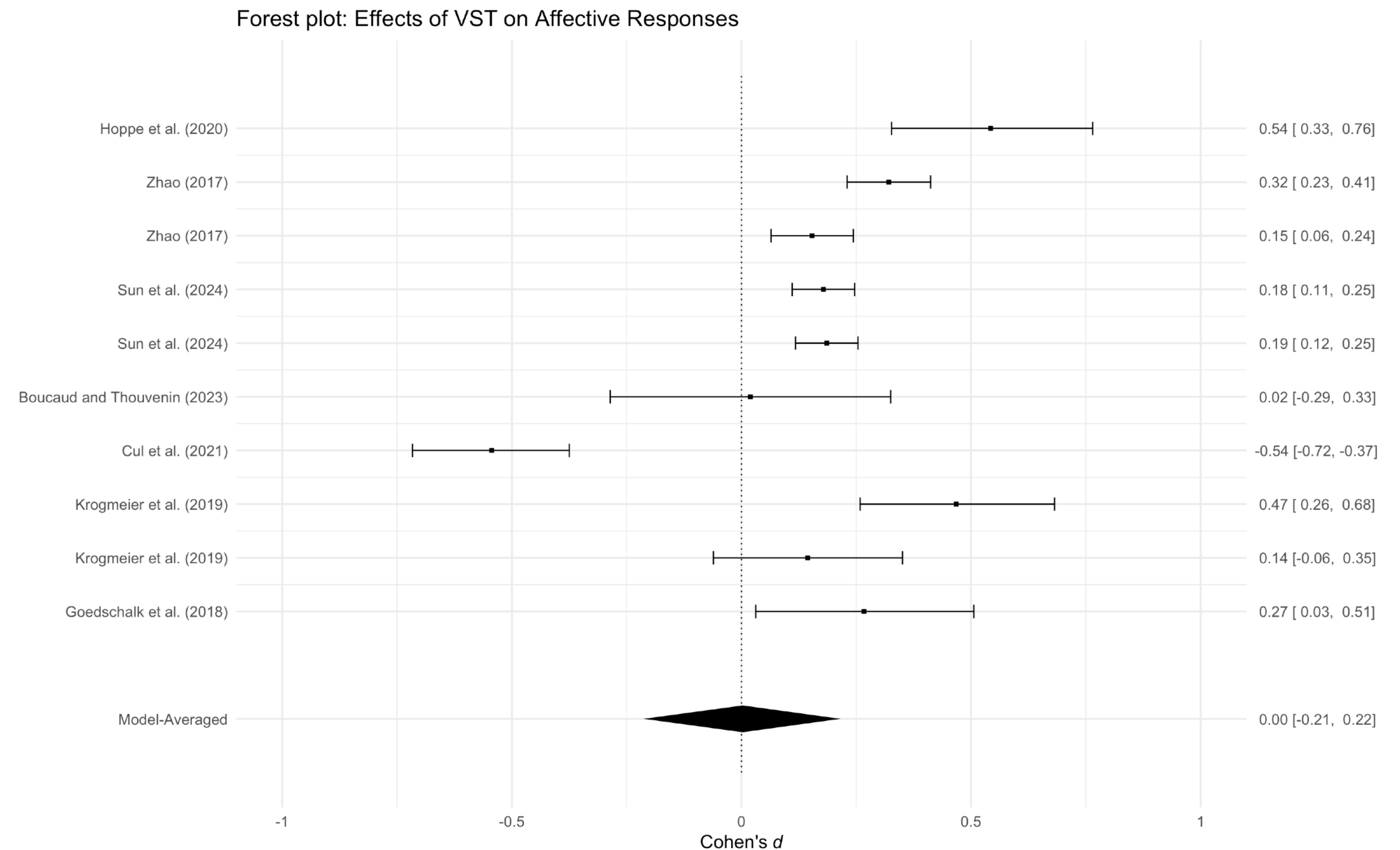
- The studies spanned from 2009 to 2024 and were sourced from both published and unpublished outlets.
- The majority of these studies were journal articles (50%) originating from various countries, including USA and Germany.
- The sample sizes of these studies ranged from 19 to 75 participants, all of which were laboratory experiments.
- Social interactions took place in various contexts, including shopping, dance lessons, ultimatum games, cooperative games, doctor-patient interactions, and nightclubs.
- Of all the effect sizes analyzed ($k = 48$), 21 (43.75%) pertained to perceptual responses, 10 (20.83%) were related to affective responses, 9 (18.75%) to behavioral outcomes, and 8 (16.67%) were associated with physiological measures.

Overall Average Effect Size

- Contrary to predictions, we found **no consistent pattern of VST effects** on interpersonal communication outcomes.

(a) Affective responses

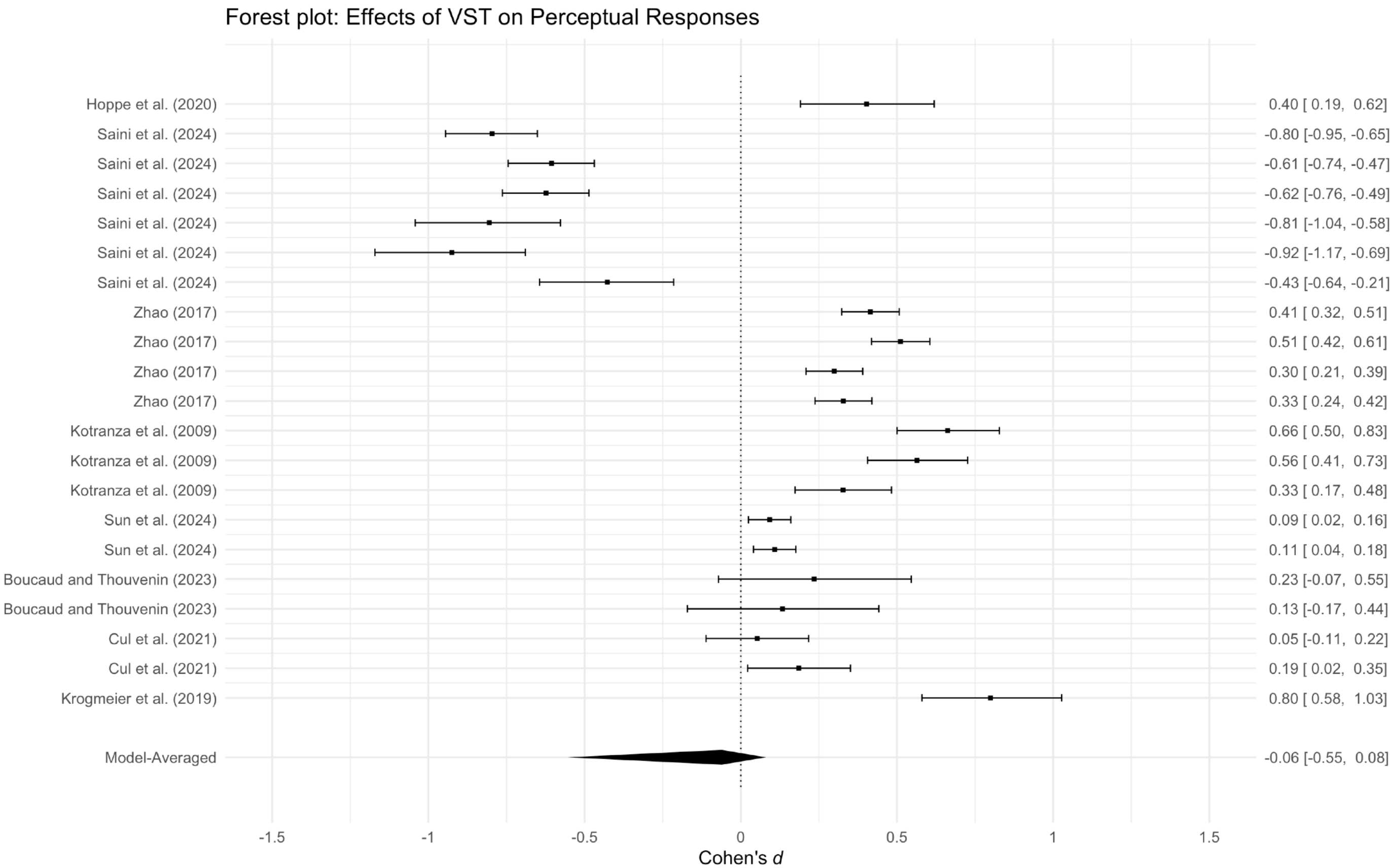
- a moderate evidence for VST's affective effects ($BF = 0.236$), with a mean model-averaged estimate of Cohen's $d = 0.003$, 95% CI $[-0.213, 0.216]$



Overall Average Effect Size

(b) Perceptual responses

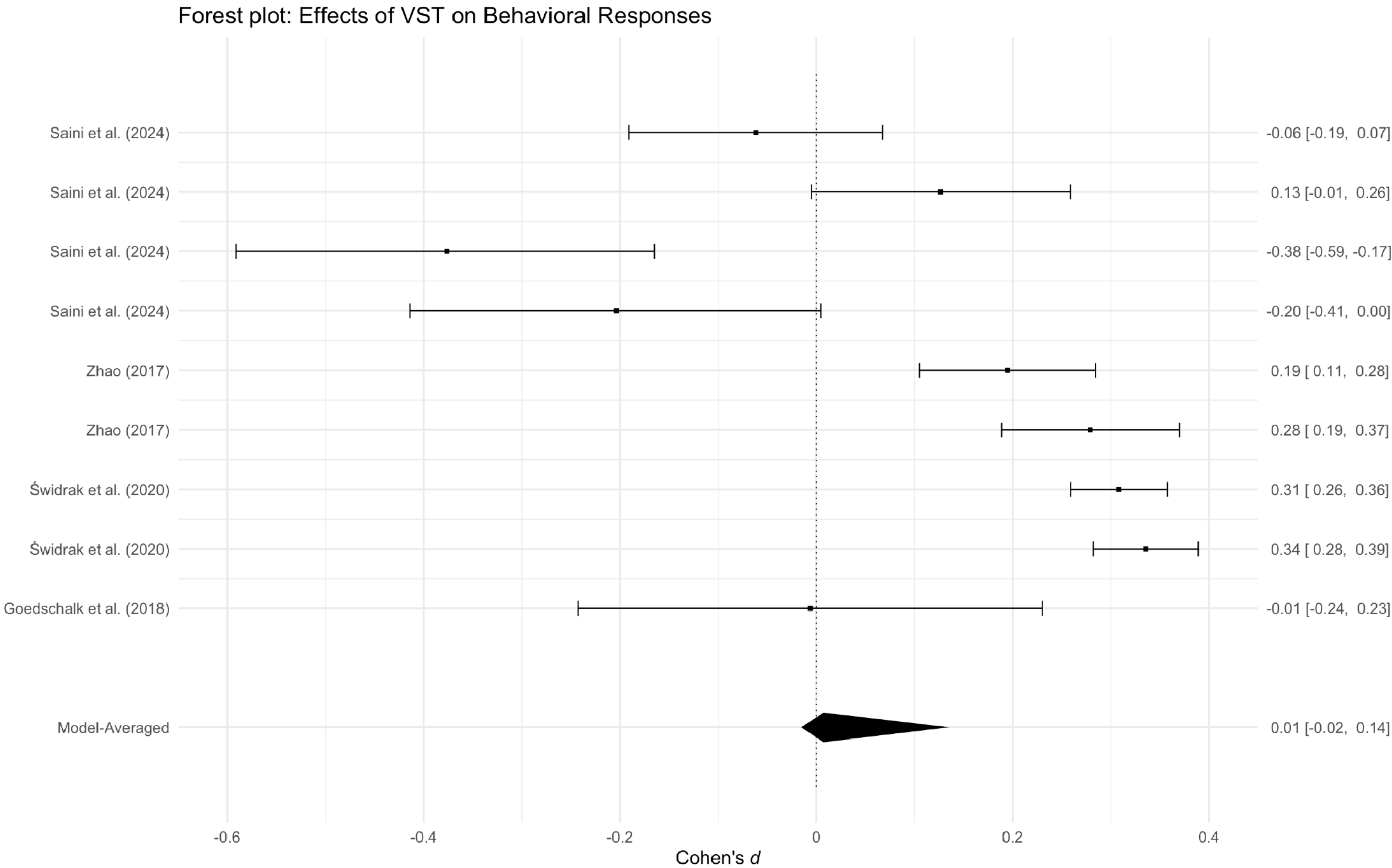
- an weak evidence for VST’s perceptual effects (BF = 0.337), with a mean model-averaged estimate of Cohen's $d = -0.061$, 95% CI [-0.554, 0.081]



Overall Average Effect Size

(c) Behavioral responses

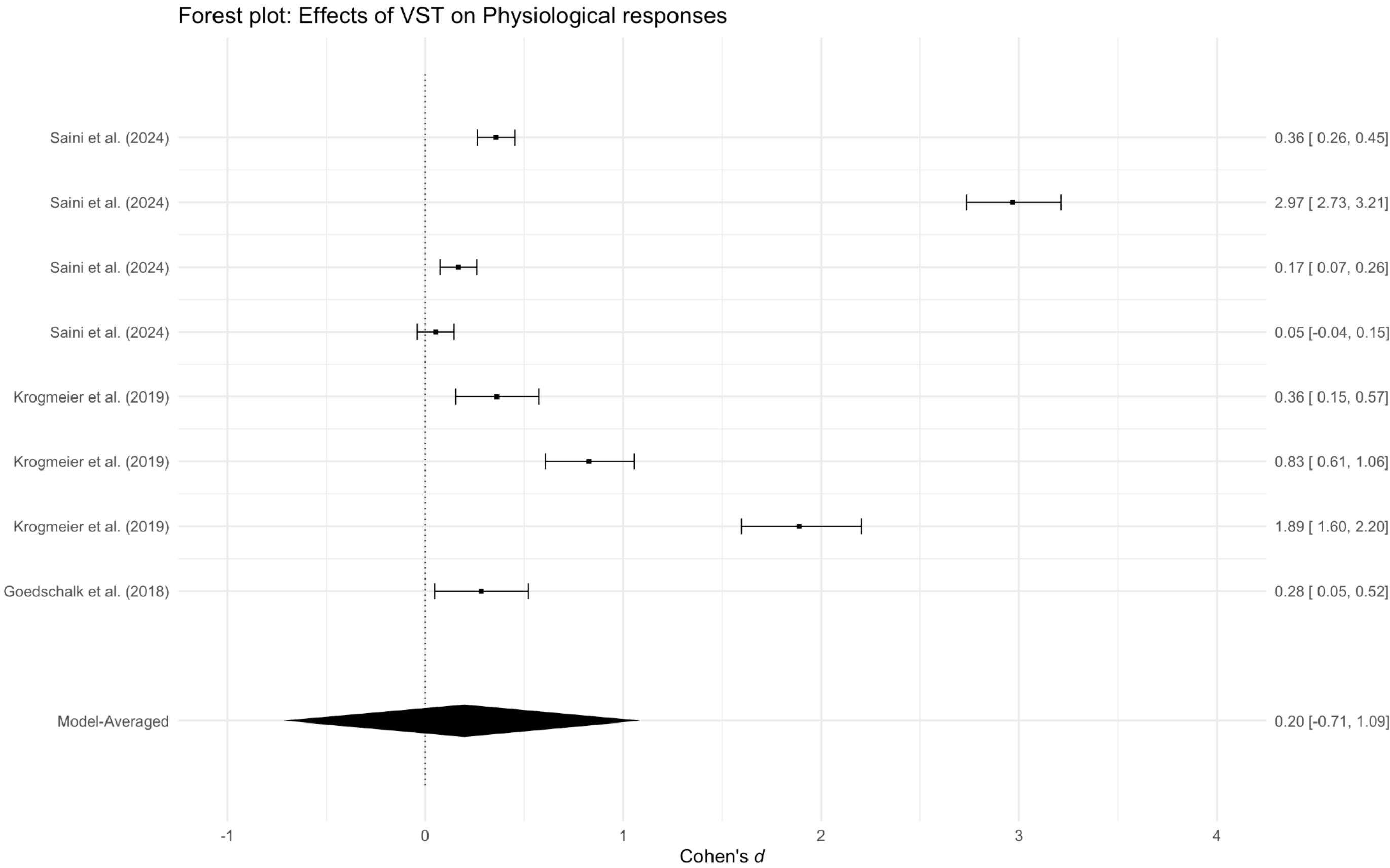
- a moderate evidence for VST’s behavioral effects (BF = 0.134), with a mean model-averaged estimate of Cohen's $d = 0.007$, 95% CI [-0.015, 0.136]



Overall Average Effect Size

(d) Physiological responses

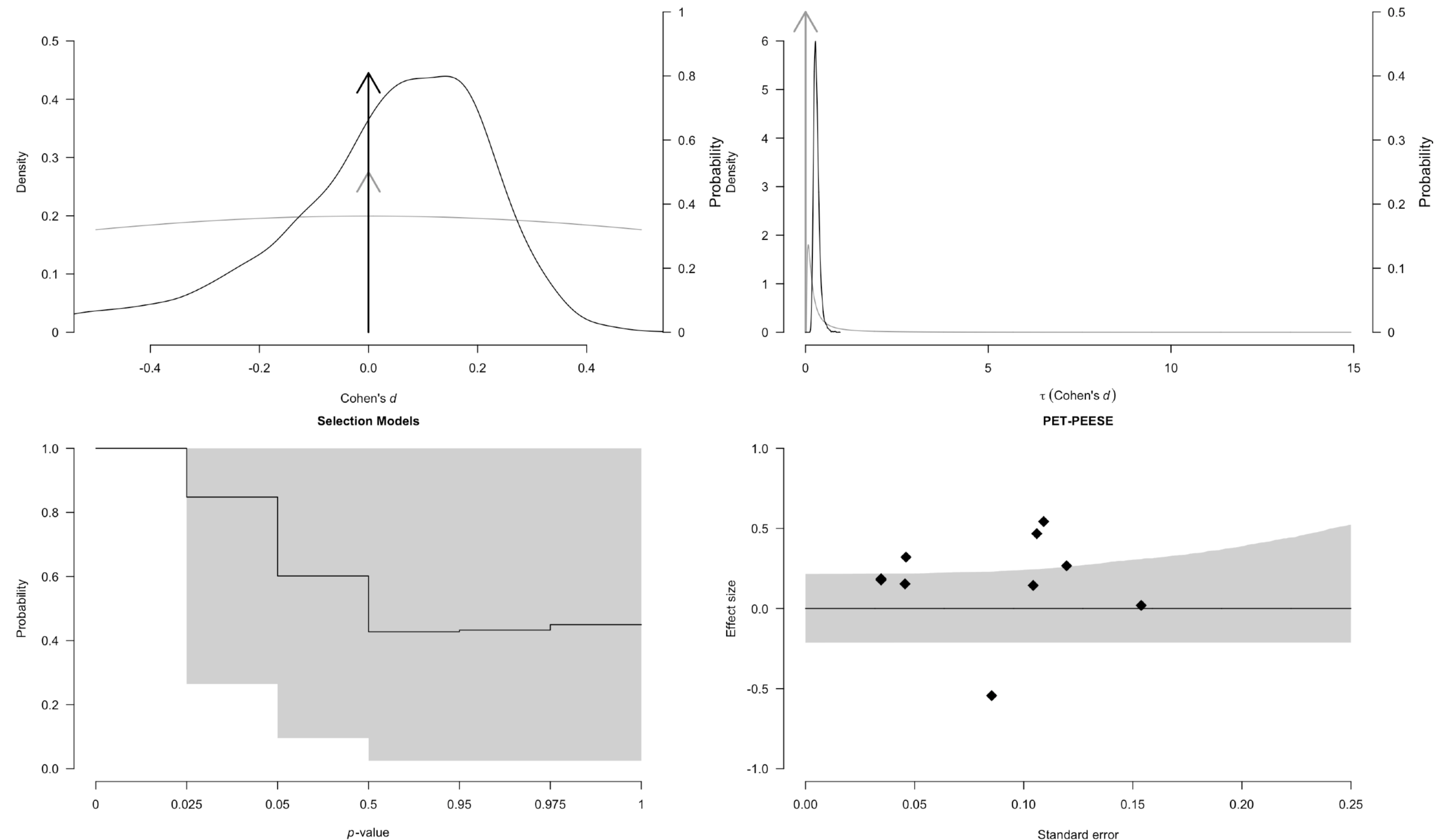
- an weak evidence for VST’s physiological effects (BF = 1.08), with a mean model-averaged estimate of Cohen's $d = 0.197$, 95% CI [-0.714, 1.087]



Heterogeneity and Publication Bias

(a) Affective responses

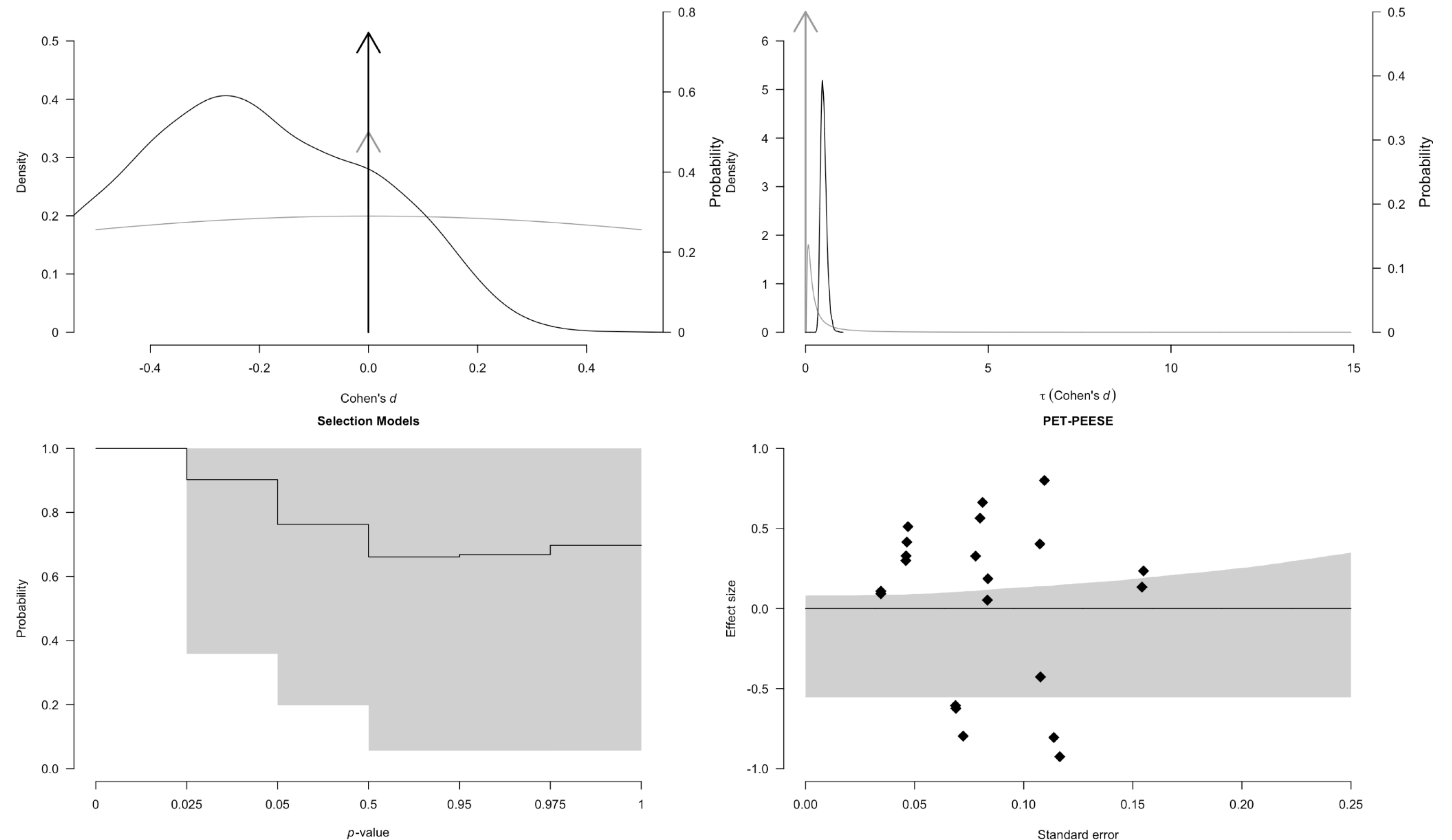
- a strong evidence in favor of the heterogeneity ($BF = 2.44e+19$), with mean model-averaged estimate $\tau = 0.303$, 95% CI [0.185, 0.503]
- a moderate evidence in favor of the publication bias ($BF = 5.49$)



Heterogeneity and Publication Bias

(b) Perceptual responses

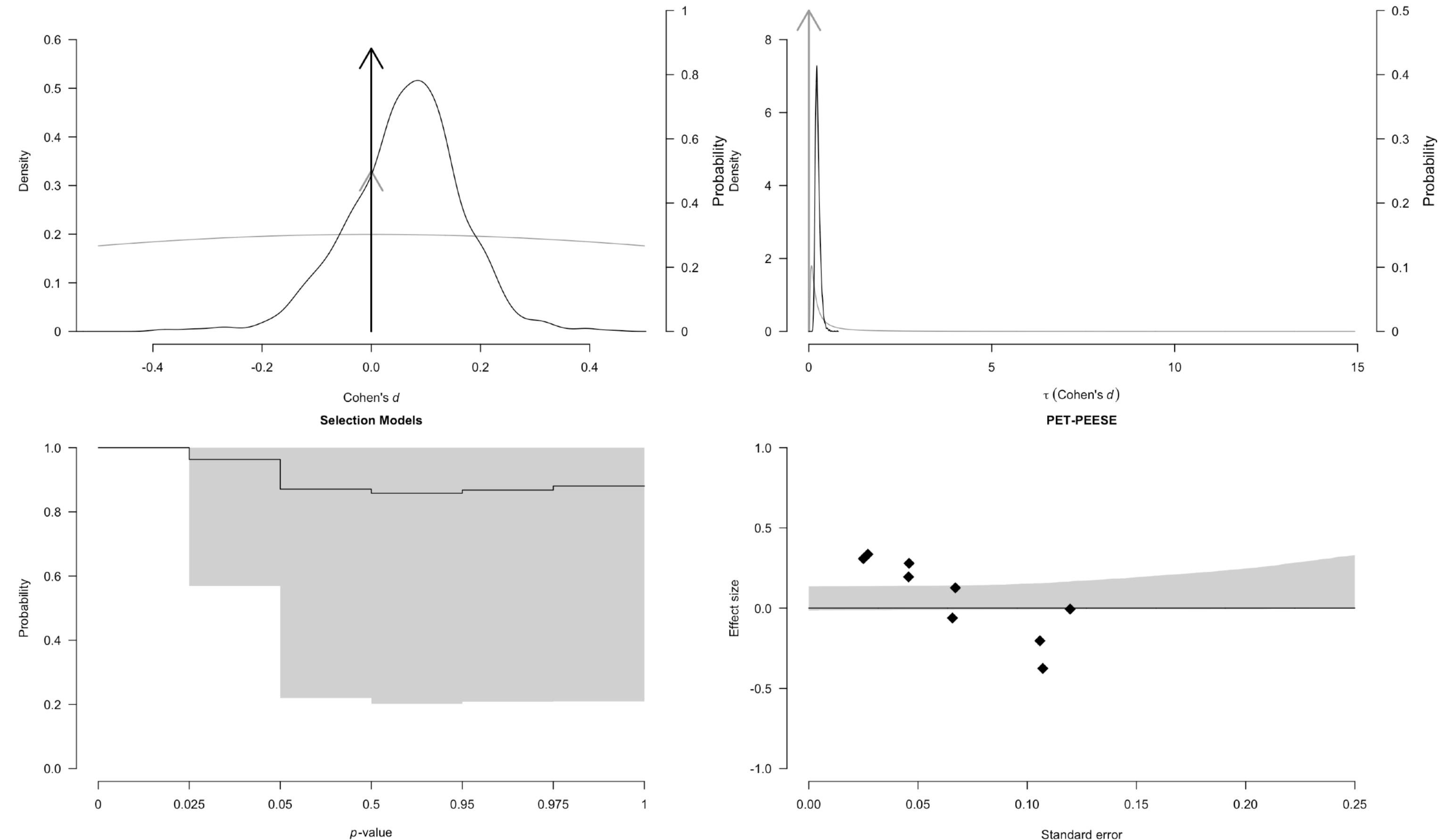
- a strong evidence in favor of the heterogeneity ($BF = 7.72e+163$), with mean model-averaged estimate $\tau = 0.496$, 95% CI [0.364, 0.685]
- an weak evidence in favor of the publication bias ($BF = 1.88$)



Heterogeneity and Publication Bias

(c) Behavioral responses

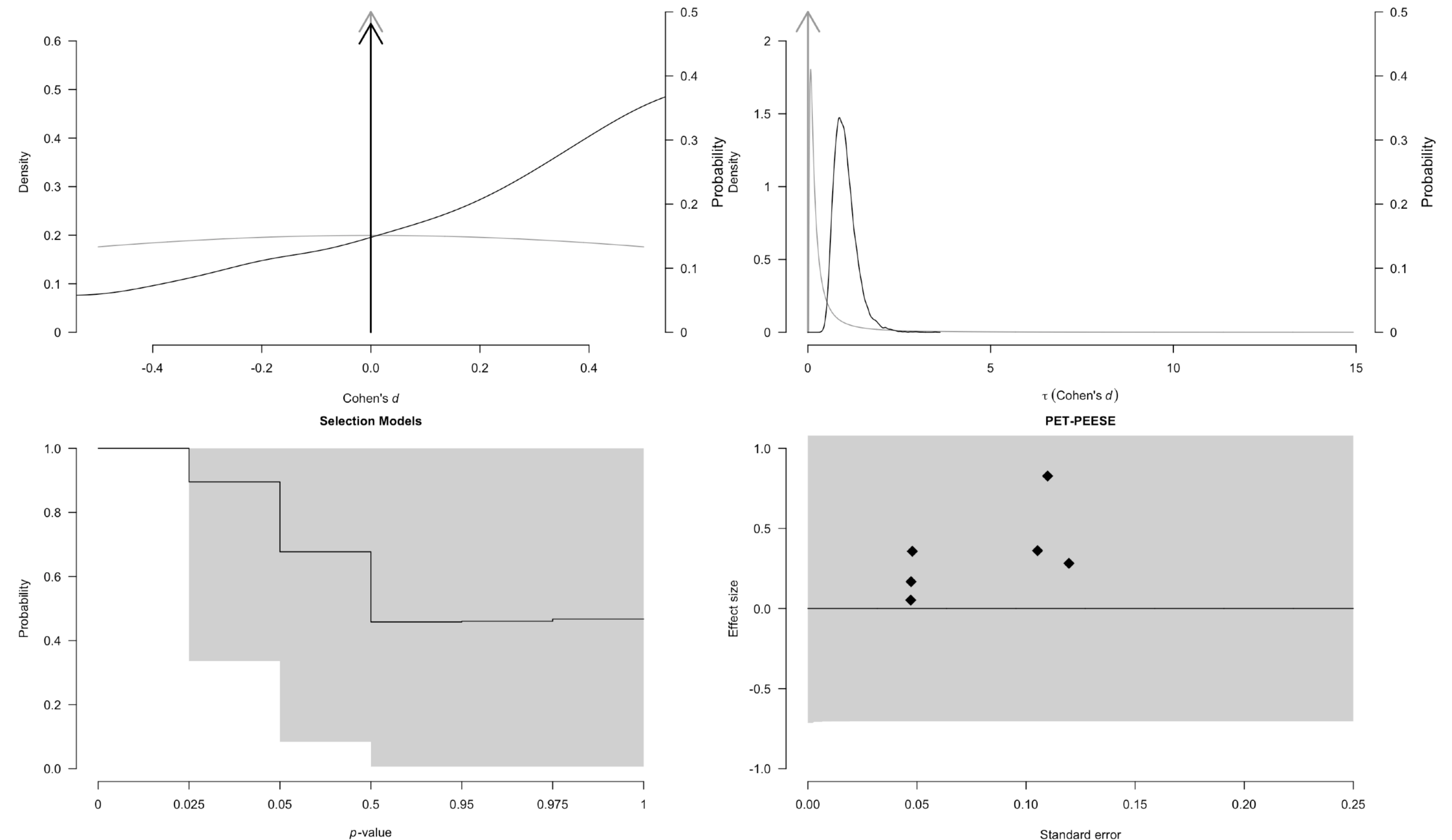
- a strong evidence in favor of the heterogeneity ($BF = 3.76e+19$), with mean model-averaged estimate $\tau = 0.244$, 95% CI [0.148, 0.403]
- an weak evidence in favor of the publication bias ($BF = 0.732$)



Heterogeneity and Publication Bias

(d) Physiological responses

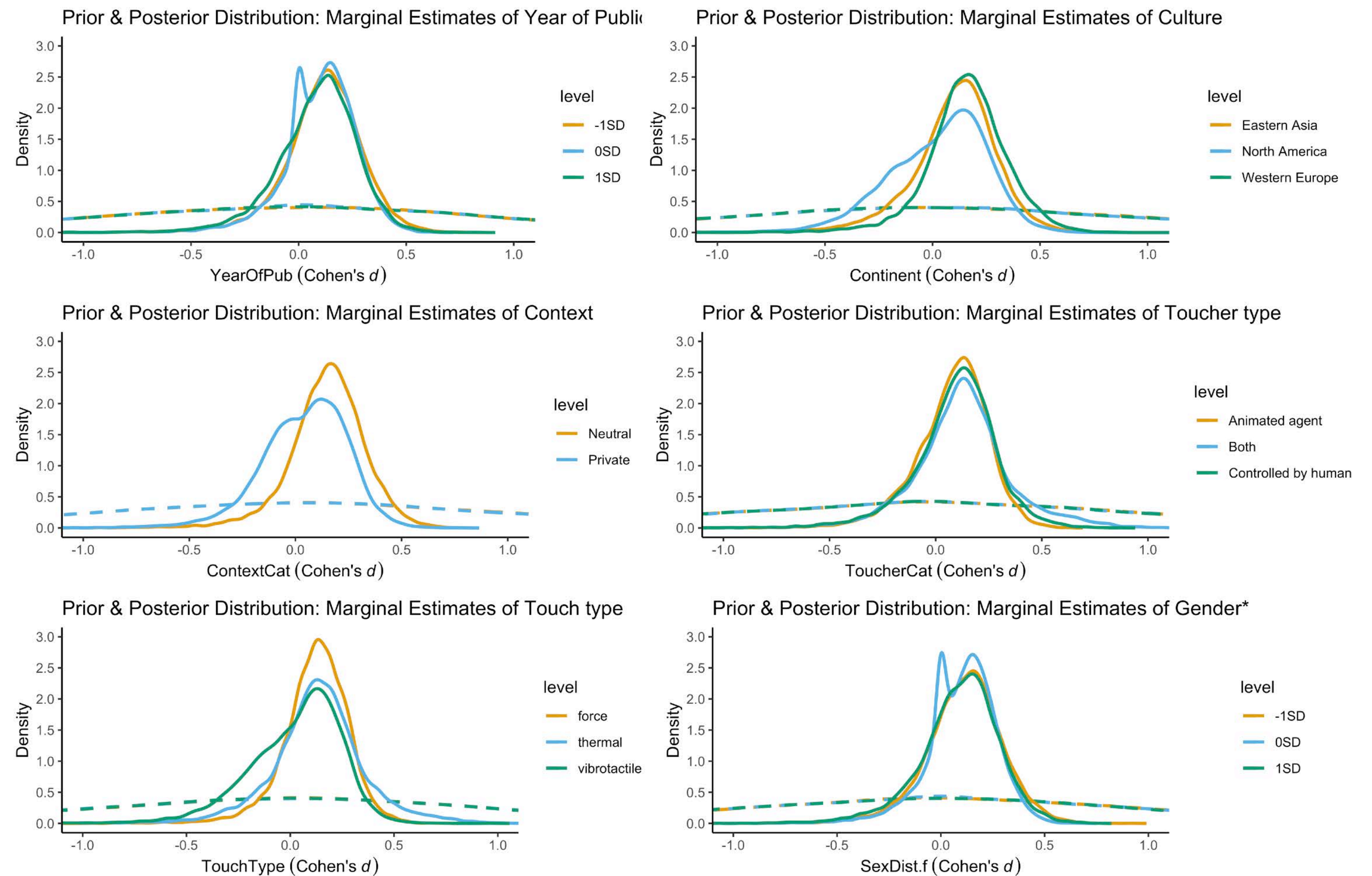
- a strong evidence in favor of the heterogeneity ($BF = 1.51e+183$), with mean model-averaged estimate $\tau = 1.027$, 95% CI [0.580, 1.809]
- a moderate evidence in favor of the publication bias ($BF = 3.97$)



Moderator Analysis

(a) Affective responses¹⁾

- an weak evidence for VST's affective effects ($BF = 0.26$), with a mean model-averaged estimate of Cohen's $d = 0.022$, 95% CI $[-0.032, 0.278]$.

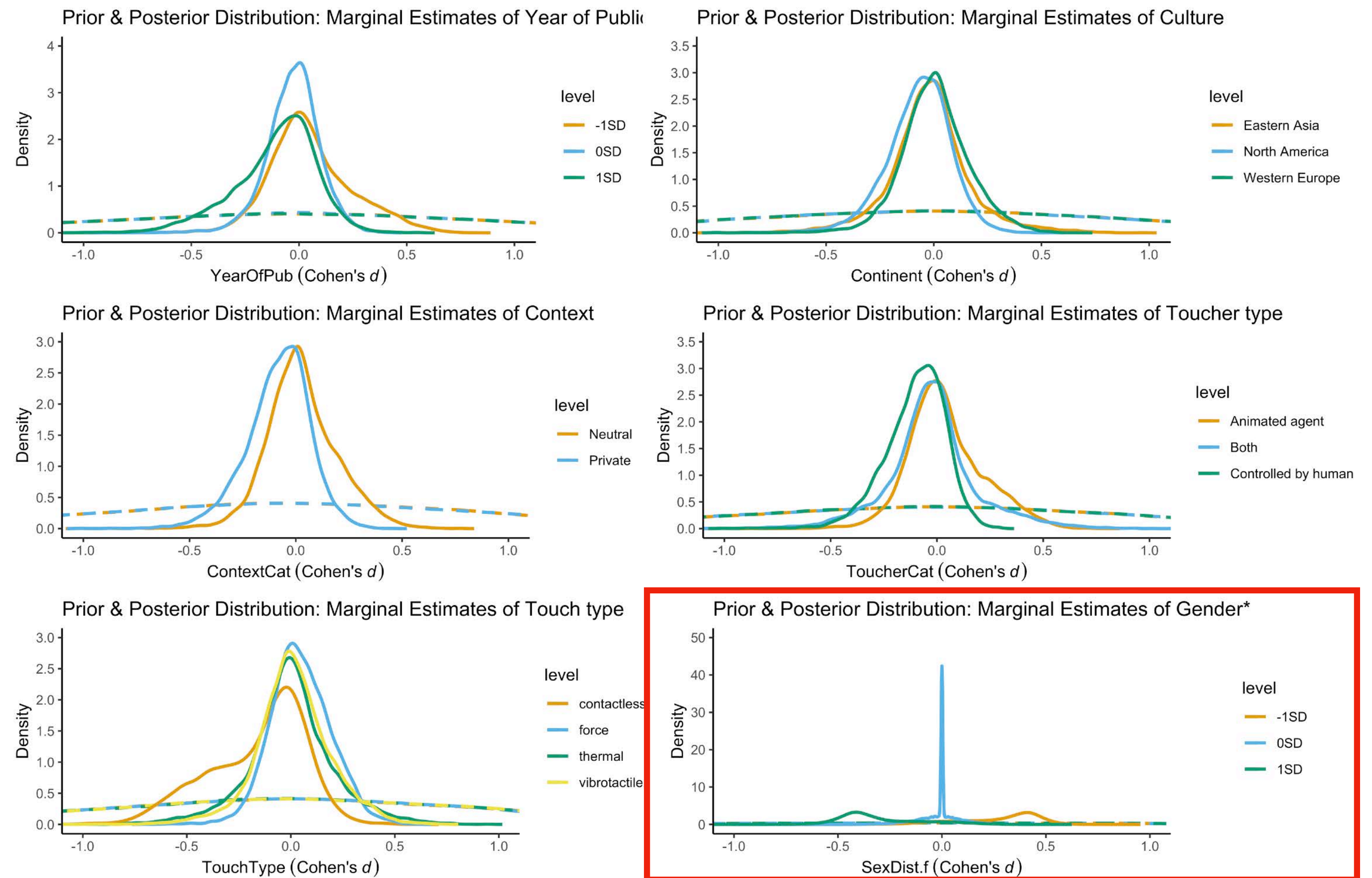


¹⁾ Due to incomplete reporting of relevant information, the participant's mean age was excluded from the analysis.

Moderator Analysis

(b) Perceptual responses¹⁾

- an weak evidence for VST's perceptual effects ($BF = 0.11$), with a mean model-averaged estimate of Cohen's $d = -0.001$, 95% CI $[-0.054, 0.030]$.
- a moderate evidence in favor of the gender variable among the moderators included ($BF = 3.38$)

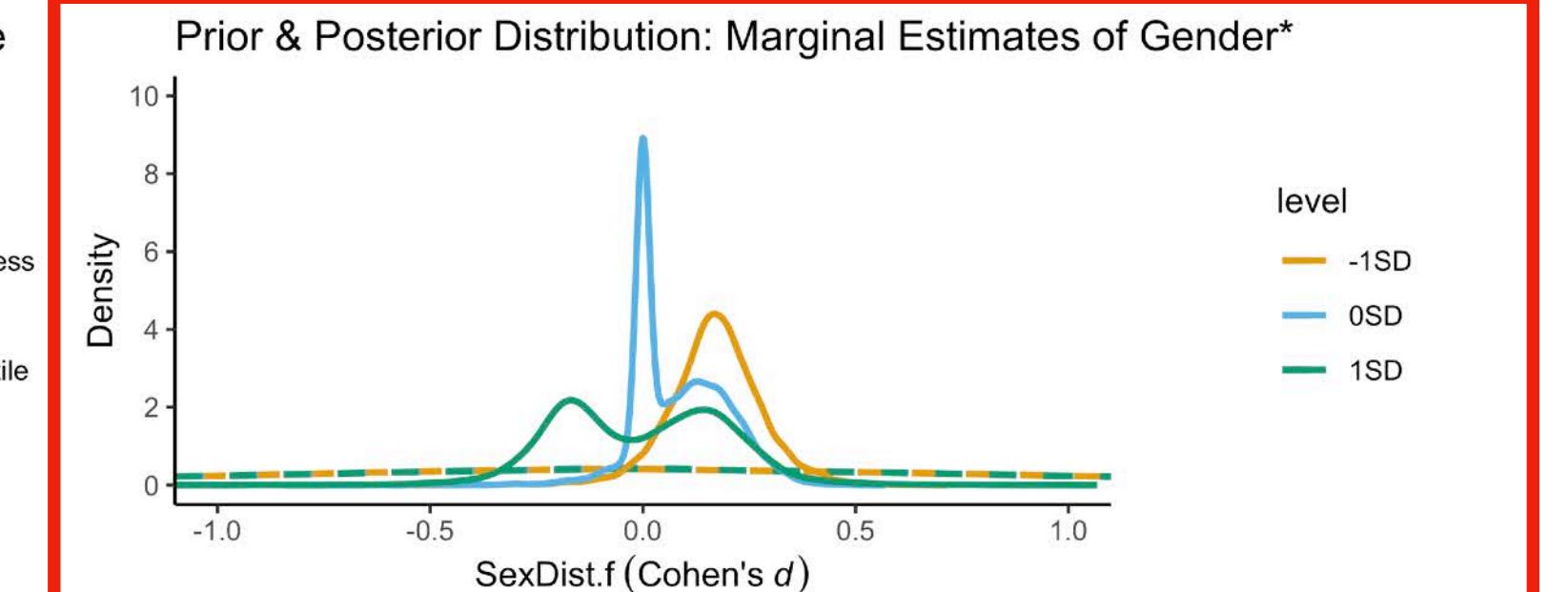
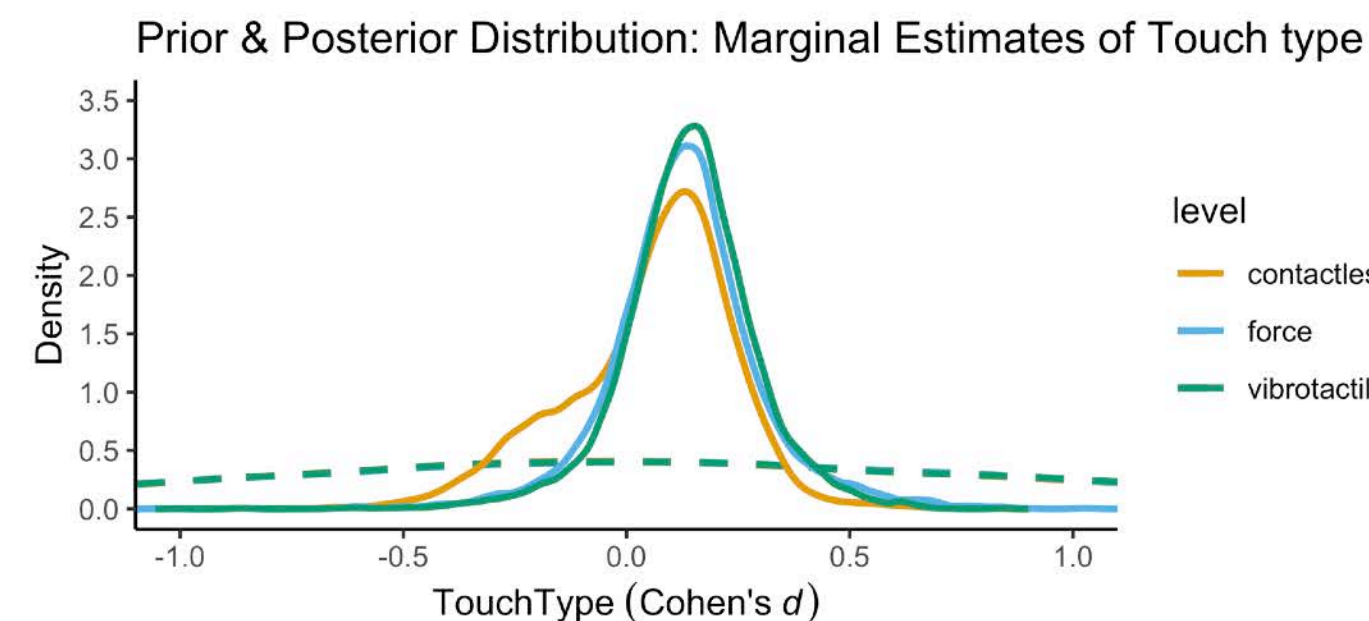
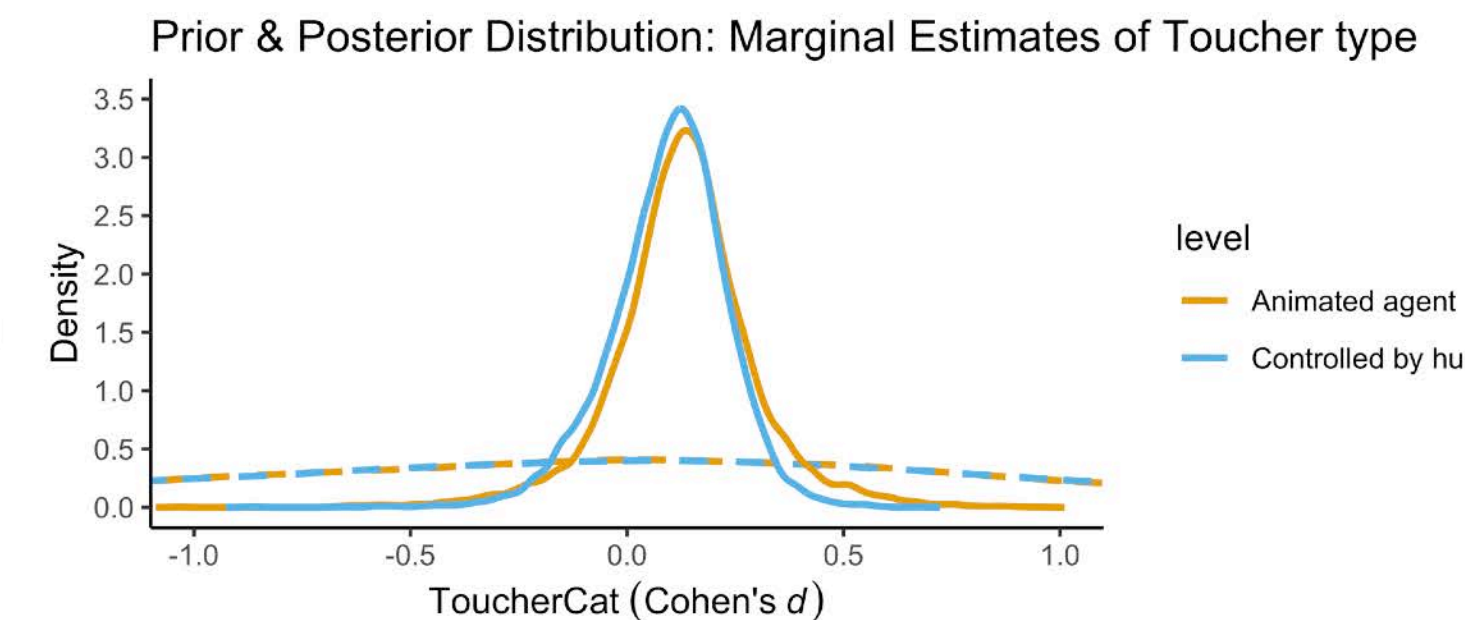
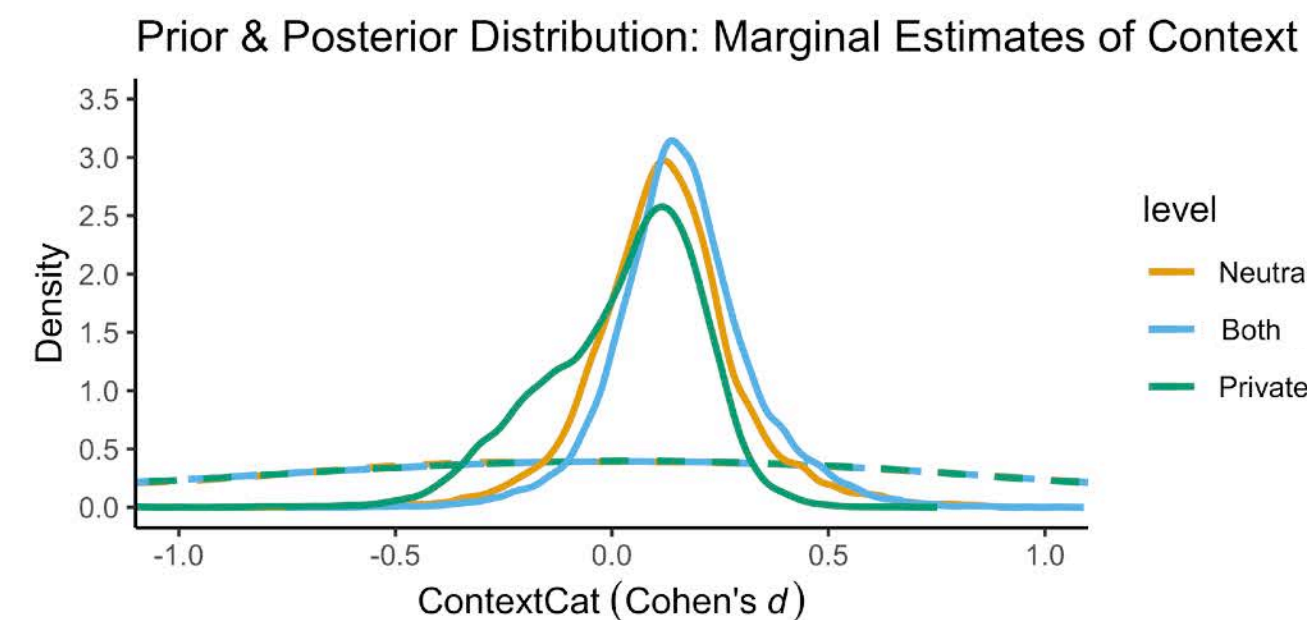
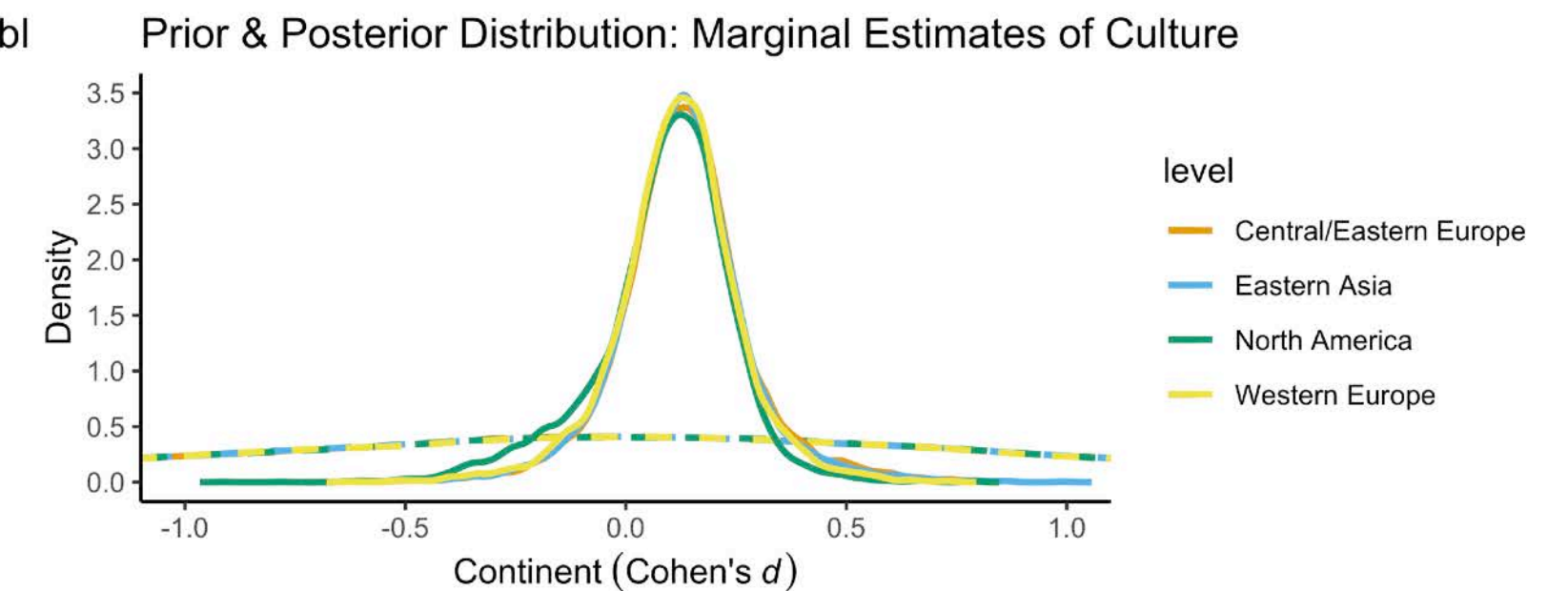
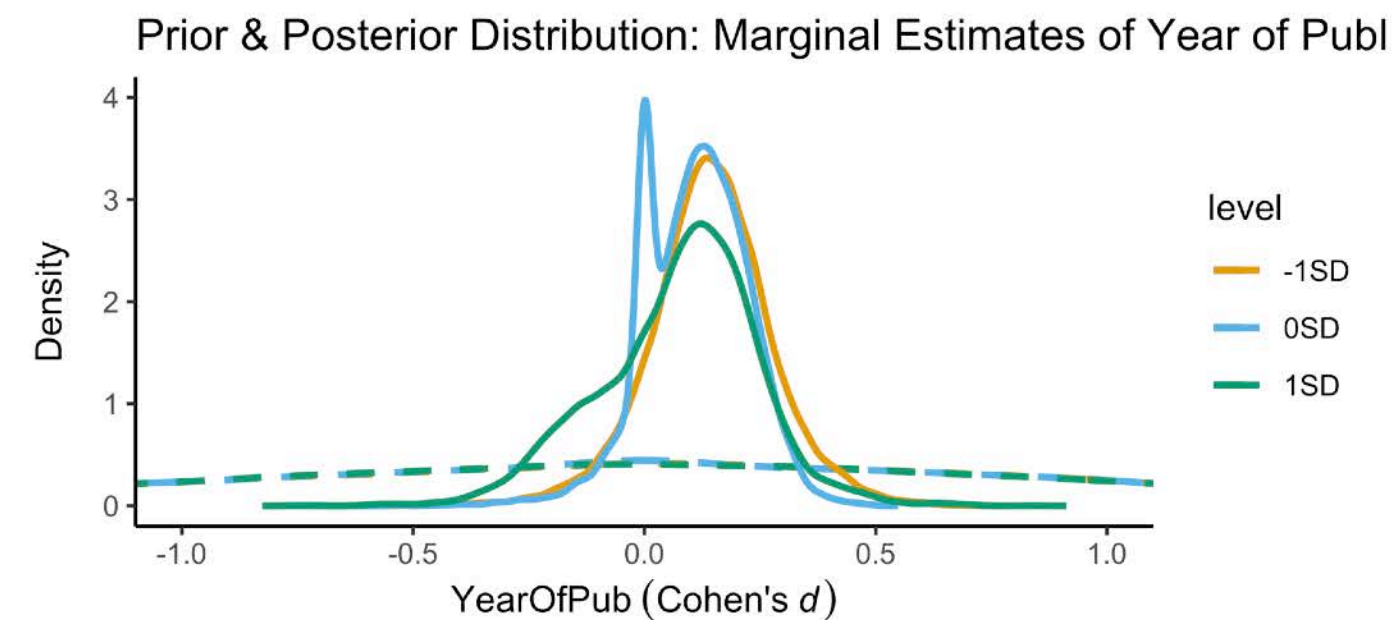


¹⁾ Due to incomplete reporting of relevant information, the participant's mean age was excluded from the analysis.

Moderator Analysis

(c) Behavioral responses¹⁾

- an weak evidence for VST's behavioral effects ($BF = 0.24$), with a mean model-averaged estimate of Cohen's $d = 0.022$, 95% CI [0.000, 0.233].
- an weak evidence in favor of the gender variable among the moderators included ($BF = 1.394$)

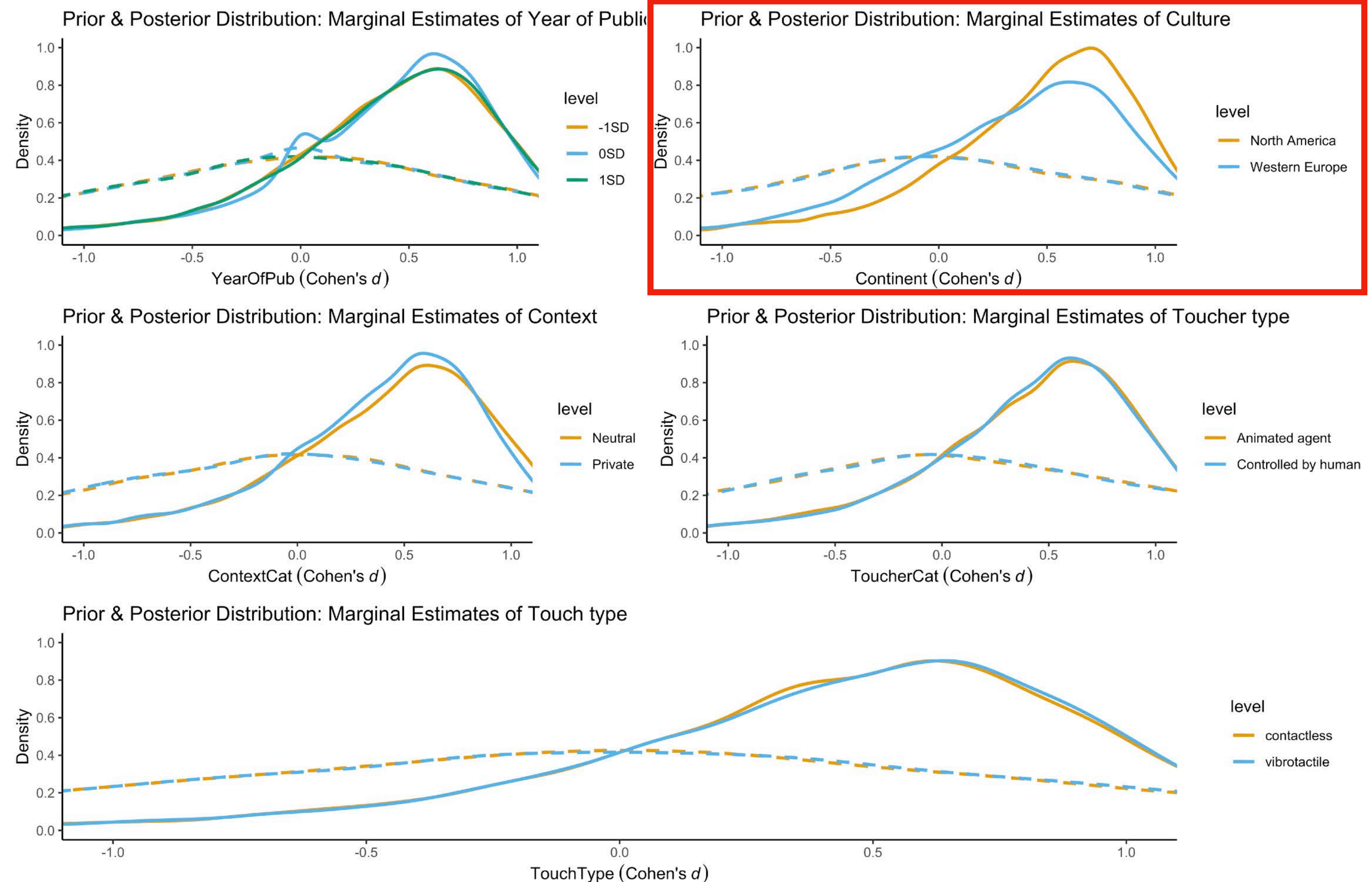


¹⁾ Due to incomplete reporting of relevant information, the participant's mean age was excluded from the analysis.

Moderator Analysis

(d) Physiological responses¹⁾

- an weak evidence for VST's physiological effects ($BF = 1.12$), with a mean model-averaged estimate of Cohen's $d = 0.232$, 95% CI $[-0.557, 1.132]$.
- an weak evidence in favor of the cultural background variable among the moderators included ($BF = 1.06$)



¹⁾ Due to incomplete reporting of relevant information, the participant's gender distribution and mean age was excluded from the analysis.

- Our meta-analysis reviews the effects of VST on interpersonal communication and highlights key gaps in knowledge that will facilitate a better understanding of human communication. We detected **no consistent evidence supporting the effects of VST in IVEs** on affective, perceptual, behavioral, and physiological communication outcomes.
- These findings imply the **complexity of mediated social touch interactions in IVEs**, indicating the necessity for a more comprehensive investigation into the intricate relationships among various parameters. As explained by Jakubiak & Feeney (2017), the process by which receiving affectionate touch is emotionally interpreted by the recipient involves traversing a *relational-cognitive pathway*, which is influenced by diverse personal, situational, relational, and cultural moderators. These factors may result in VST not always yielding expected effects.

- Despite the overall lack of effect of VST on interpersonal communication outcomes, our meta-regression analysis revealed the **potential for gender differences in perceptual and behavioral outcomes**. According to these findings, it can be inferred that men tend to evaluate immersive social interaction experiences and partners more positively, engaging in greater amount of interaction or exhibiting more pro-social behaviors. This aligns with studies indicating that gender influences the frequency of social touch and intimacy in VEs. Based on related studies, tactile stimulation may potentially enhance task motivation for males, with touch effects on male compliance influenced by factors such as perceived partner value and social status, rather than touch alone. Considering touch's fundamental role in real-world compliance, power relations, and affective phenomena (Hertenstein et al., 2006), these findings indicate that **virtual social interaction outcomes involve complex interactions beyond touch stimuli presence**.
- However, given the absence of strong and consistent evidence for gender differences in current study, it highlights the importance of further exploring in VST research. Investigating the nuanced interplay between gender and VST can provide valuable insights into understanding how individuals interact and perceive interactions within IVEs.

- Furthermore, although significant results were not obtained in this study, all instances of social touch effects contrary to expectations (i.e., negative effects), as reviewed by the authors, stemmed from experimental studies conducted within intimate contexts (e.g., Cui et al., 2021; Saini et al., 2024). Whether touch is received from a familiar or unfamiliar individual is an important concept modulating the effects of touch in the real world. Given that most touch occurrences in real-life happening within intimate context, SVT within familiar interaction settings in experimental studies may yield adverse effects. **Therefore, it emphasizes the importance of carefully designing the interaction context in which exposure to SVT occurs in future research, based on existing experiential evidence.**

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